Laser Beam Tracker & Profiler

Itamar Aloni

Abstract

This report presents the development of a webcam-based Laser Beam Profiler designed to track, measure, and analyze laser beam characteristics in real time using a standard camera. The system uses image-processing algorithms to detect and track the laser beam, determine its center, and estimate its width along both spatial axes.

The measurement process consists of five main steps:

1. Background subtraction
2. Recursive Region of Interest (ROI) detection
3. Subpixel beam center estimation
4. Beam width calculation using the $1/e^2$ intensity criterion
5. Gaussian fitting (R^2)

The algorithms were initially developed and tested using simulated Gaussian beams with known parameters, allowing their accuracy to be evaluated under different beam sizes and noise conditions. The system was then implemented using a standard webcam (Microsoft LifeCam HD-3000) to demonstrate real-time beam tracking and analysis.

The results demonstrate the system's ability to detect, track, and characterize Gaussian beam profiles, providing a foundation for further improvements in measurement accuracy and spatial calibration.

1. Beam Generator

1.1 Electric Field and Optical Intensity

For an Ideal circular Gaussian beam, the complex field envelope may be written as:

$$E(x,y) = E_0 \exp\left(-\frac{r^2}{w^2}\right) e^{i\phi(x,y)}, \qquad r^2 = (x-x_0)^2 + (y-y_0)^2$$

Here (x_0, y_0) is the beam center and w is the beam radius. For monochromatic wave in a homogeneous, nonmagnetic medium, the time averaged intensity is:

$$I = \frac{1}{2} n \varepsilon_0 c |E|^2$$

Therefore, we can rewrite the Intensity equation as:

$$\boxed{I(x,y) = I_0 \exp\left[-2\frac{(x-x_0)^2 + (y-y_0)^2}{w^2}\right]}$$

This intensity distribution is the reference model used throughout this project.

1.2 Beam Radius and Diameter

A Gaussian beam radius is defined as the point which its intensity falls to $1/e^2 \approx 13.5\%$ of its peak:

$$I(r = w) = I_0 e^{-2}, \qquad D_{1/e^2} = 2w.$$

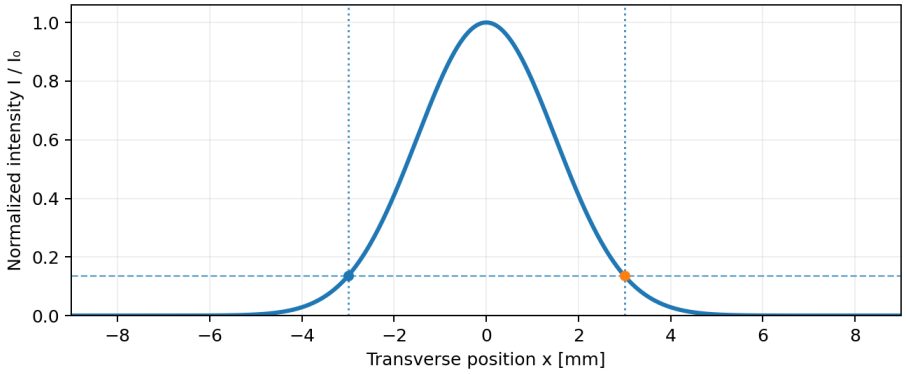


Figure 1. Normalized transverse intensity profile of an ideal Gaussian beam, with the 1/e² intensity crossings marked.

1.3 Synthetic Beam Generation

To verify the algorithms before using a camera, a gaussian image is sampled on a known Cartesian grid. A constant background B and additive Gaussian noise N are included:

$$I_{\text{sim}}(x,y) = \max[0,\ I(x,y) + B + N(x,y)], \qquad N \sim \mathcal{N}(0, \sigma^2).$$

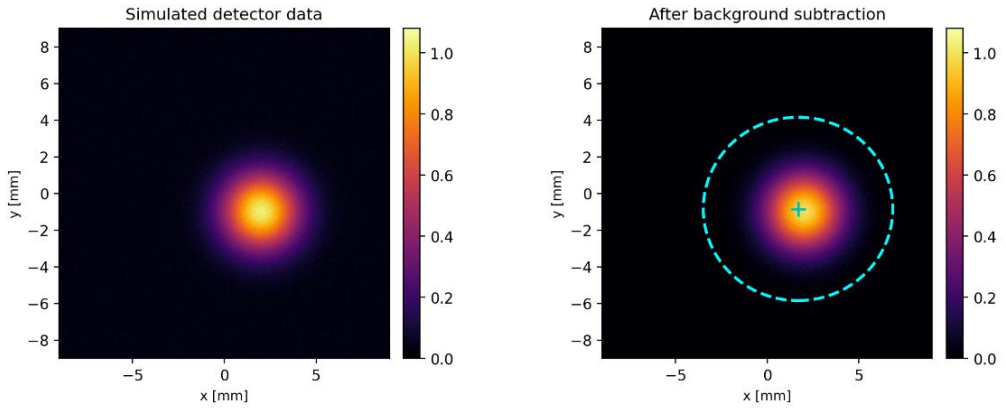


Figure 2: Simulated Gaussian beam on a 601 × 601 grid spanning −9 to 9 mm, with a known center of (1, −3) mm and a radius of 1 mm. The beam is shown before (left) and after (right) background subtraction, with an initial background level of 0.03 and Gaussian noise of σ = 0.02 (normalized intensity).

2. Region of Interest (ROI) Detection

2.1 Background Correction and peak detection

For camera measurements, the system averages 30 frames to estimate the background intensity at each pixel. The background is then subtracted from each new color frame, with negative values clipped to zero. The highest corrected color-channel value is selected for each pixel to create a single intensity map. Finally, Gaussian smoothing is applied to reduce noise and prevent isolated high-intensity pixels from being incorrectly identified as the beam's peak.

2.2 Peak Constrained Recursive Search

Processing the full frame is unnecessary once the beam region is found. The image is divided into a grid, and the mean intensity of each cell is calculated. The score of a candidate 2 x 2 group of adjacent cells is:

$$S_{ij} = \mu_{ij} + \mu_{i+1,j} + \mu_{i,j+1} + \mu_{i+1,j+1}.$$

Only candidate groups containing the detected peak are considered. The highest-scoring group becomes the search region for the next iteration. The current implementation first uses an 8x8 grid and then a 4x4 grid inside the selected region. This reduces the analysis area while keeping the strongest detected peak inside the final ROI.

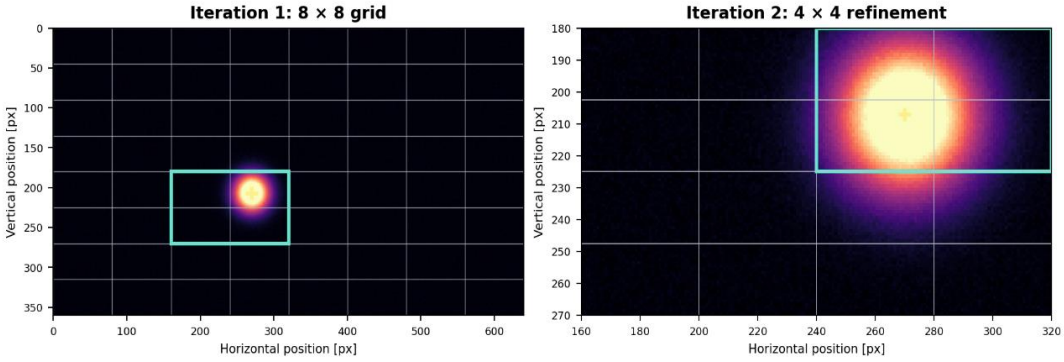


Figure 3. Two-stage recursive ROI selection. The highlighted rectangle is retained at each stage; the cross marks the peak.

3. Beam Center Estimation

The selected ROI provides an initial peak position, not the true beam center. The corrected image is smoothed to identify a connected signal region. Only the component containing the detected peak is retained. The unsmoothed, non-negative intensity values then act as centroid weights:

$$\hat{x}_c = \frac{\sum_{x,y} x\, W(x,y)}{\sum_{x,y} W(x,y)}, \qquad \hat{y}_c = \frac{\sum_{x,y} y\, W(x,y)}{\sum_{x,y} W(x,y)}$$

The centroid is re-calculated in a local window until the position changes by less than the chosen tolerance or the iteration limit reached. This produces sub-pixel coordinates, greatly improving preciseness.

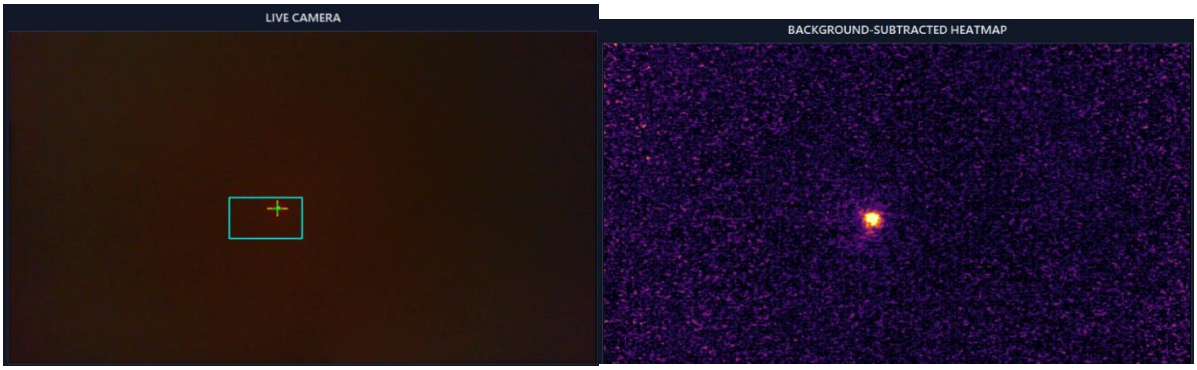


Figure 4. Laser spot detection in the camera image and the corresponding background-subtracted intensity map.

4. Beam Width measurement

The image is reduced to horizontal and vertical intensity profiles by averaging along the perpendicular axis. After residual-background removal and mild one-dimensional smoothing, the peak is located near the estimated center. For each axis, the algorithm finds the two crossings of $I_{\max}/e^2$ and interpolates between adjacent samples:

$$\boxed{w_x = \frac{x_R - x_L}{2}, \qquad w_y = \frac{y_R - y_L}{2}}.$$

The calculated values represent beam radii, not diameters. Averaging helps reduce noise, while interpolation improves the accuracy of the measurements between pixels. If the beam is too wide for the analyzed region, the measurement is rejected.

5. Gaussian Fitting

An independent estimate of each radius is obtained by fitting the measured profile to

$$I_{\text{fit}}(u) = A\exp\left[-2\left(\frac{u-u_c}{w_{\text{fit}}}\right)^2\right] + B$$

where A, u_c, w_{fit}, and B are fitted parameters. Least-squares optimization uses the full sampled profile rather than just the two threshold crossings. The coefficient of determination, $R^2 = 1 - \sum(I - I_{\text{fit}})^2 / \sum(I - \bar{I})^2$, describes how well the measured shape follows a Gaussian model. For an ideal Gaussian beam, the fitted and direct $1/e^2$ radii should agree. A discrepancy may indicate noise, non-Gaussian tails, saturation, or background errors. A high R^2 alone does not prove physically accurate camera calibration.

6. Experimental Validation

6.1 synthetic Width Tests

The unchanged width estimator was tested on six known radii, with 100 independent noisy realizations per radius ($\sigma = 0.02$, $I_0 = 1$). The true center was supplied to isolate width-estimation error from center-localization errors.

Table 1. Synthetic Gaussian beam width validation across six known radii.

True radius	Mean estimated X	Mean estimated Y	Both axes within 5%
0.40	0.4041	0.4044	87 / 100
0.60	0.6023	0.6003	99 / 100
1.00	0.9991	0.9986	100 / 100
1.50	1.4980	1.4975	100 / 100
2.50	2.4931	2.4939	100 / 100
3.00	2.9938	2.9924	100 / 100

At $w = 1$, a noise-free test returned approximately $w_x = 1.0020$ and $w_y = 1.0021$. An elliptical noise-free beam with known $(w_x, w_y) = (0.8, 1.5)$ returned approximately $(0.8026, 1.5012)$. The largest relative errors occur for the narrowest beams, for which sampling and smoothing have greater influence.

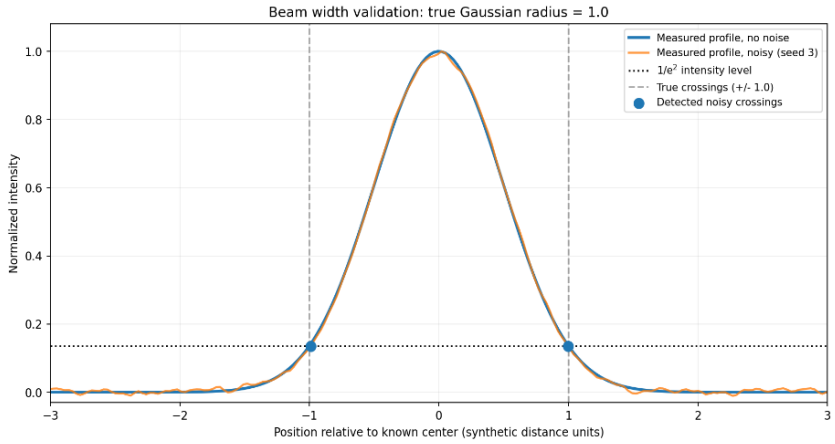


Figure 5. Validation of the 1/e² beam-radius calculation using a simulated Gaussian intensity profile.

6.2 Physical Measurements and Limitations

Webcam Tests demonstrate that the pipeline can detect a laser spot, estimate its center, and display live traverse profiles. However, webcam pixel values are not calibrated optical irradiance: automatic image processing, saturation, exposure and background conditions can alter the apparent profile. Pixel-to-millimeter conversion based on camera field of view is also approximate and requires experimental verification with a target of known size

7. Showcase

Press the Photo to open the YouTube video:

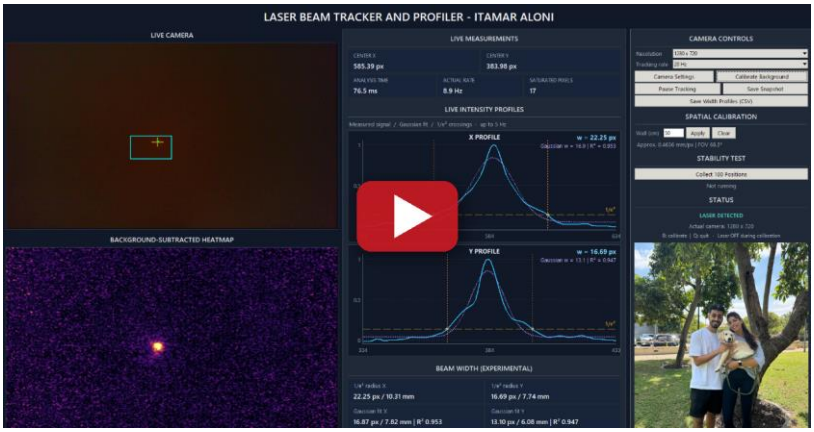


Figure 6. Laser Beam Profiler running in real time. Click the image to open the demonstration video.

8. Conclusion

The synthetic tests support the numerical correctness of the width estimator for the tested Gaussian models. They have not yet established traceable physical accuracy for the complete camera-based profiler. The next validation stage is a repeatable spatial-calibration experiment and comparison against an independent beam-size reference.